

Harrow Bay – Geophysical Survey Summary

Array area, OSS and export cable route

Sofie Lindqvist · 28 June 2024 · HB-GEO-RPT-0009

1. Survey scope

The geophysical survey was carried out between August and November 2023 with two vessels. It covered the full array area (198 km²) and a 500 m corridor along the export cable route.

Method	Purpose	Line spacing
Multibeam echosounder (MBES)	Bathymetry, seabed features	Full coverage
Side-scan sonar (SSS)	Boulders, debris, wrecks	50 m
Magnetometer	Ferrous objects, UXO	15 m in micro-siting boxes
Sub-bottom profiler / UHR seismic	Shallow geology, buried channels	25 m

2. Bathymetry

Water depths range from 27.6 m (south-west) to 36.4 m LAT (north-east). The seabed is generally flat with sand waves and megaripples in the southern part of the array.

3. Seabed obstructions

A dense boulder field was identified north of the planned OSS location, with more than 400 boulders larger than 0.5 m per km². The OSS was moved 380 m to the south-west, and WTG 22 and WTG 31 were micro-sited by 120 m and 85 m respectively. Elsewhere, boulder density is low.

Feature	Count	Action
Boulders > 0.5 m (array)	2,874	Clearance where within 50 m of foundation or cable
Magnetic anomalies > 50 nT	312	UXO risk assessment, 41 targets for inspection
Confirmed UXO	3	Removed under marine licence, Sept 2025
Wrecks	2	500 m archaeological exclusion zone
Existing cables crossed by export route	1 (telecom)	Crossing agreement in place

4. Shallow geology

UHR seismic shows a buried glacial channel crossing the north-east corner (WTG 37–40), infilled with soft clay up to 18 m thick. This was later confirmed by CPTs and is the design driver for the longest monopiles.

5. Recommendations

- Pre-construction UXO inspection of all targets within 15 m of the final positions.
- Boulder clearance campaign on the cable routes ahead of cable lay.
- Repeat MBES on the sand wave area 6 months before cable lay.